Computer Vision I

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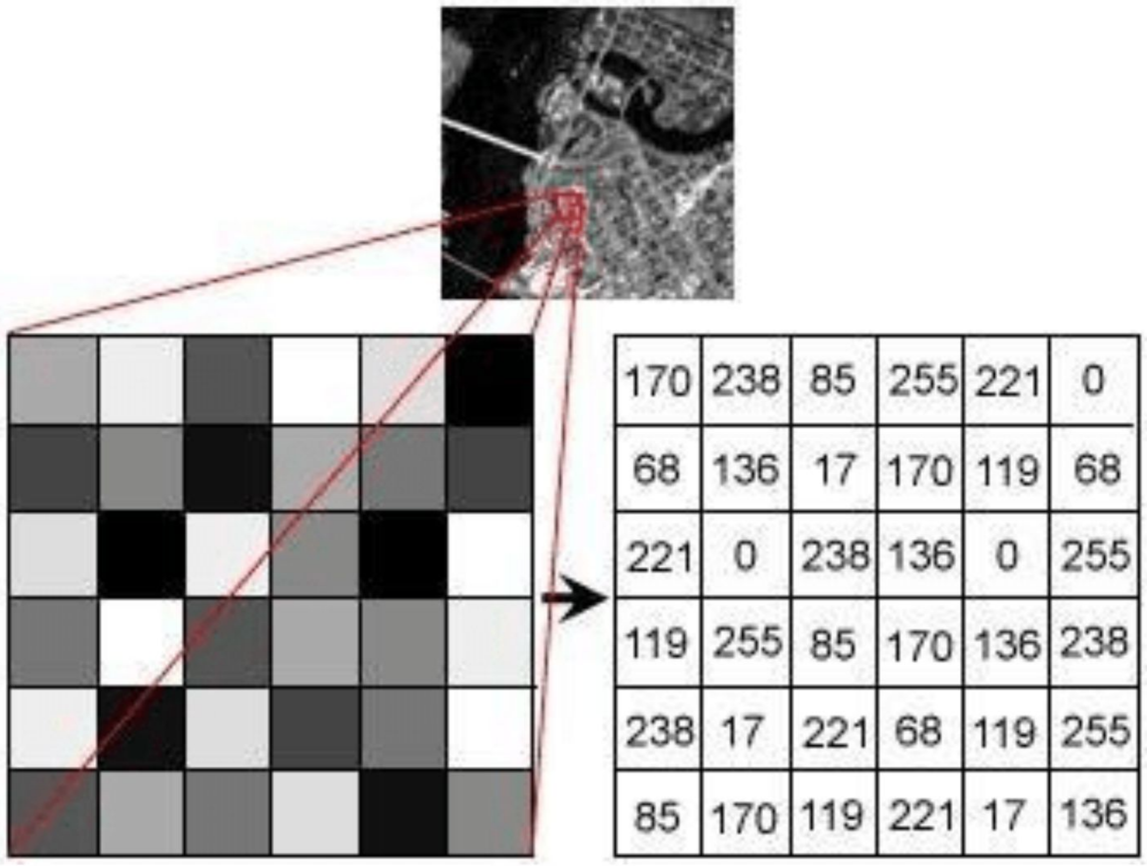
Image Representation



(1) **Pixel:** An image is a grid of squares called pixels.

(2) **Pixel Intensity**

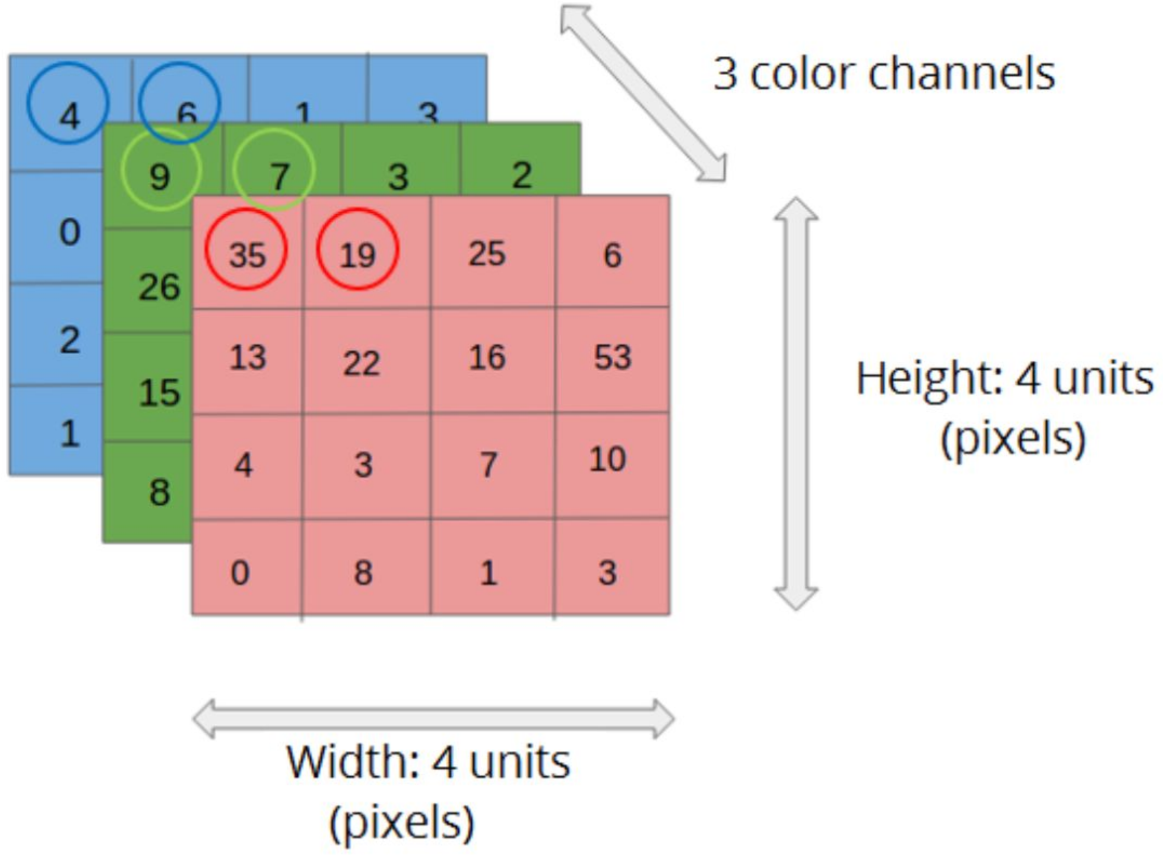
0 = Black: The absence of light.

255 = White: Maximum brightness (for an 8-bit image).

The Middle: Numbers between 0 and 255 represent shades of gray. Lower numbers are darker; higher numbers are lighter.

(3) A grayscale image is a matrix (Rows x Columns)

Image Representation



(1) A RGB image is three grayscale images stacked together.

- Layer 1 (Red Channel): Tells the computer how much red light to use.
- Layer 2 (Green Channel): Tells the computer how much green light to use.
- Layer 3 (Blue Channel): Tells the computer how much blue light to use.

(2) A RGB image is a 3D Array or tensor (Rows x Columns x Channels) .

https://datahacker.rs/image-representation

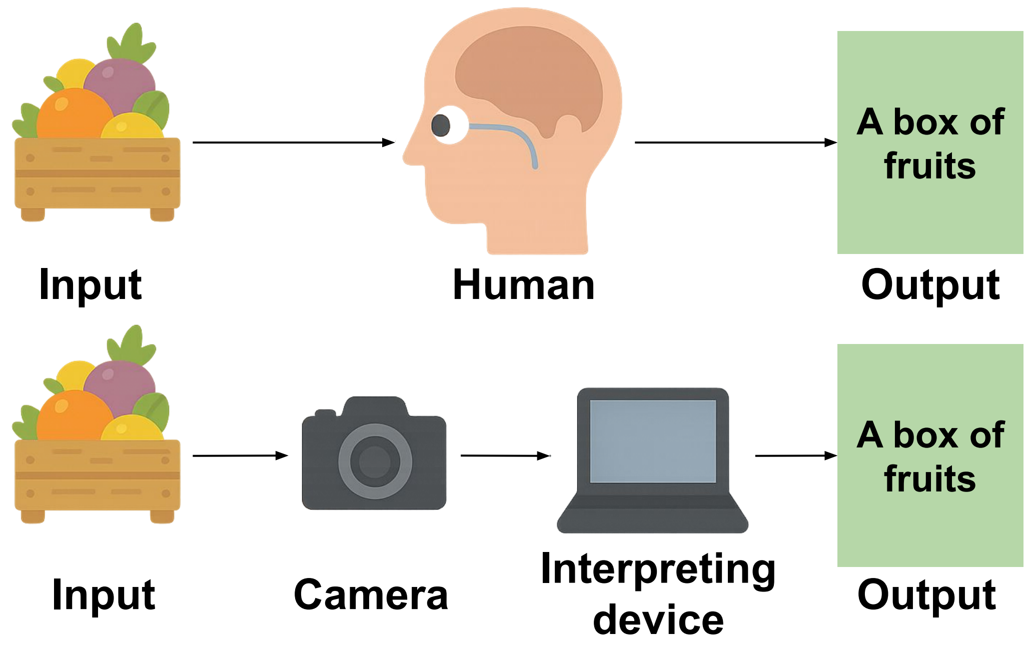
Image Processing

Raw image **Processed image**

Image processing focuses on transforming an image into another image with better quality or clearer visual information.

Common tasks: (1) image denoising, (2) image enhancement, (3) super-resolution, (4) image filtering, and (5) edge extraction.

Computer Vision



Computer vision focuses on enabling computers to understand and interpret visual information from images or videos.
Instead of producing another image, computer vision outputs semantic information or recognition results.

Common tasks:

(1) image classification

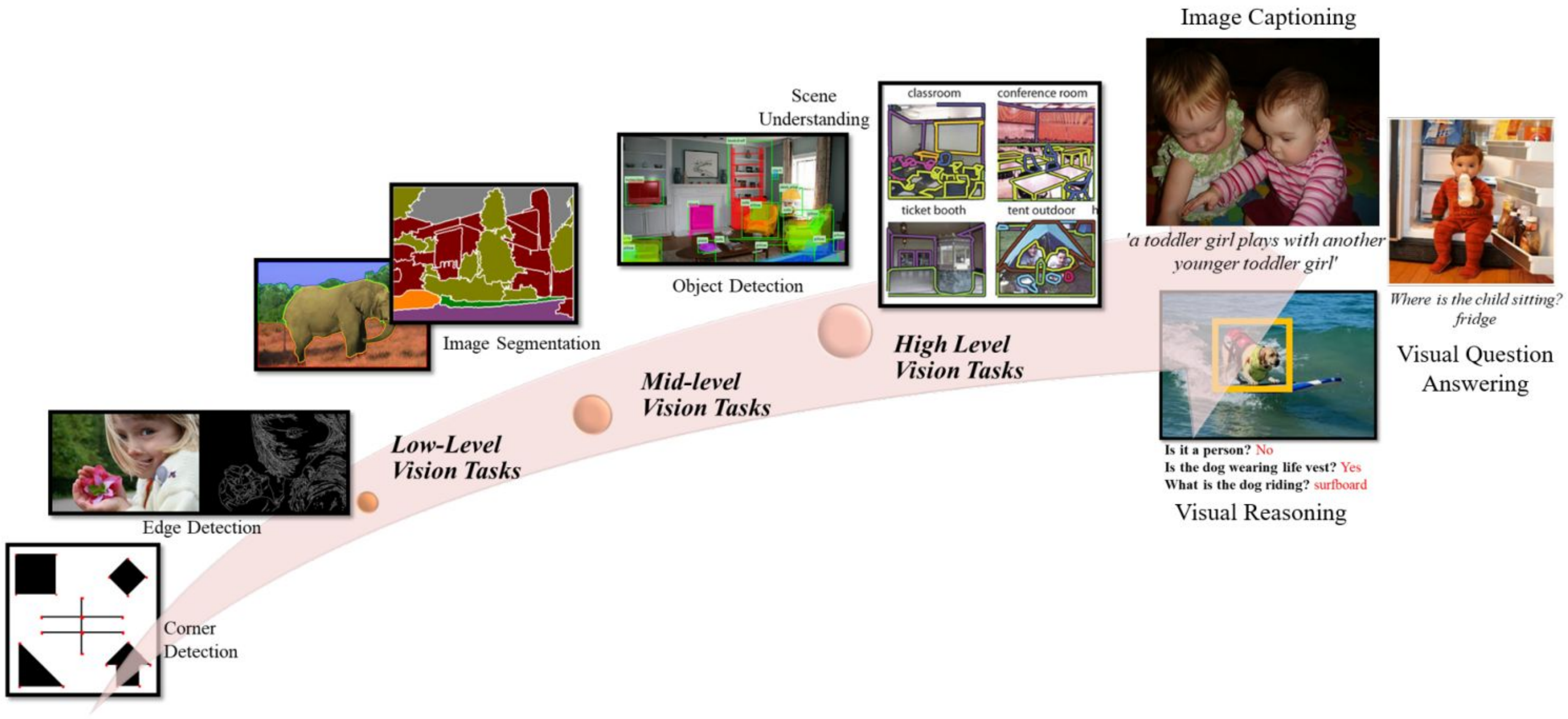
(2) object detection

(3) semantic segmentation

(4) face recognition

(5) action recognition

Image Processing vs. Computer Vision



Sharif, N., Nadeem, U., Shah, S. A. A., Bennamoun, M., & Liu, W. (2020). Vision to language: Methods, metrics and datasets. In *Machine Learning Paradigms: Advances in Deep Learning-based Technological Applications* (pp. 9-62). Cham: Springer International Publishing.

CV - Categorised by Input Modality

- **2D image**
  - Classification
  - Object detection
  - Semantic segmentation
  - Pose estimation
- **Video (2D over time)**
  - Action recognition
  - Tracking
  - Video segmentation
- **3D Data (depth maps, point clouds, meshes, voxels, multi-view images)**
  - 3D reconstruction
  - Depth estimation
- **Multimodal Input**
  - VQA
  - Text to image generation

2D Image - Classification

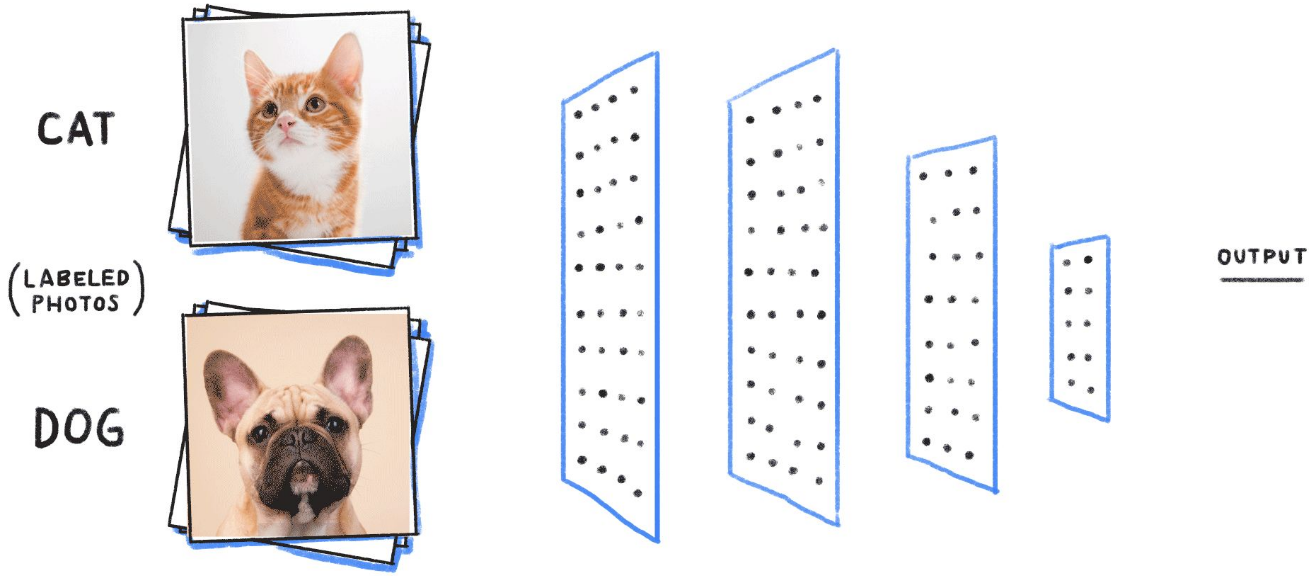
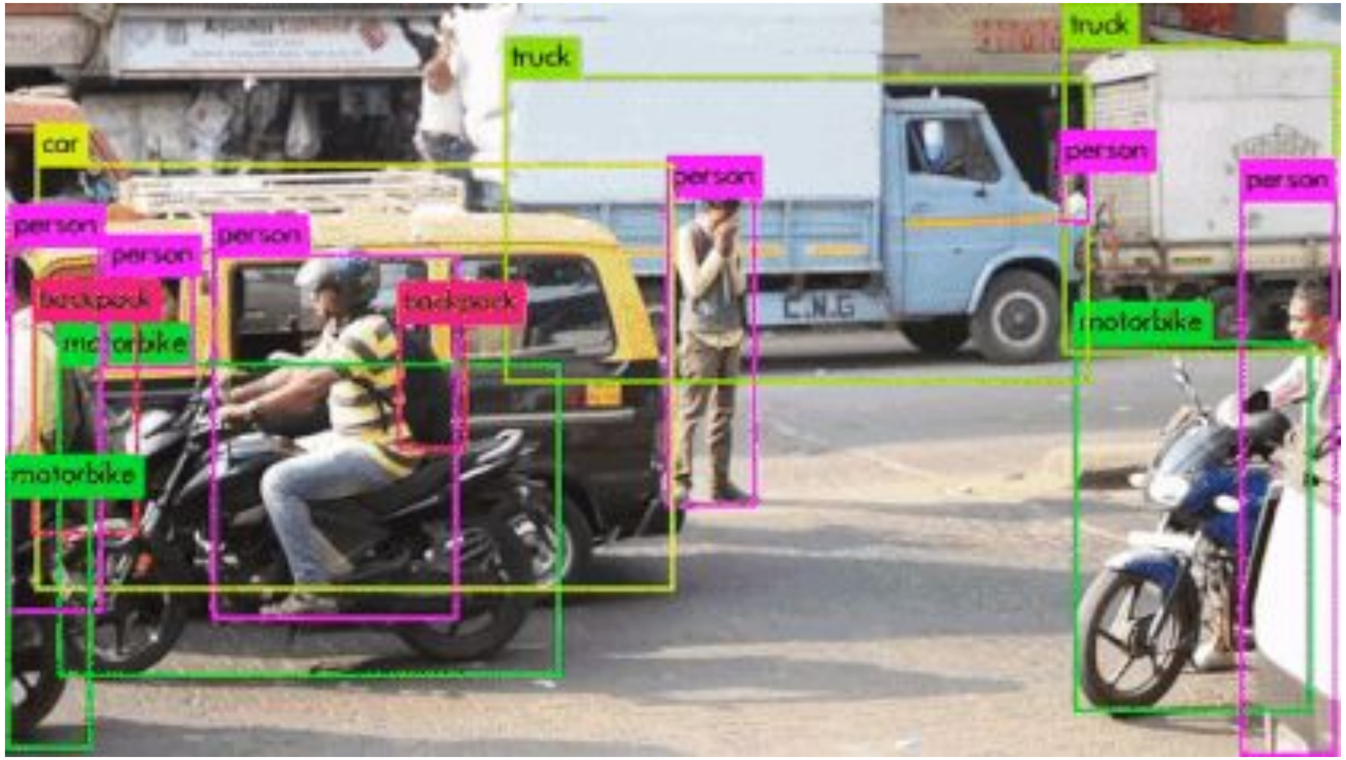


Image classification aims to assign an input image to one of the predefined categories.

- **Input:** Image (e.g., 224×224×3)
- **Output:** Probability distribution over classes (e.g., [0.8, 0.2])

https://www.kaggle.com/code/arbazkhan971/image-classification-using-cnn-94-accuracy

2D Image - Object Detection

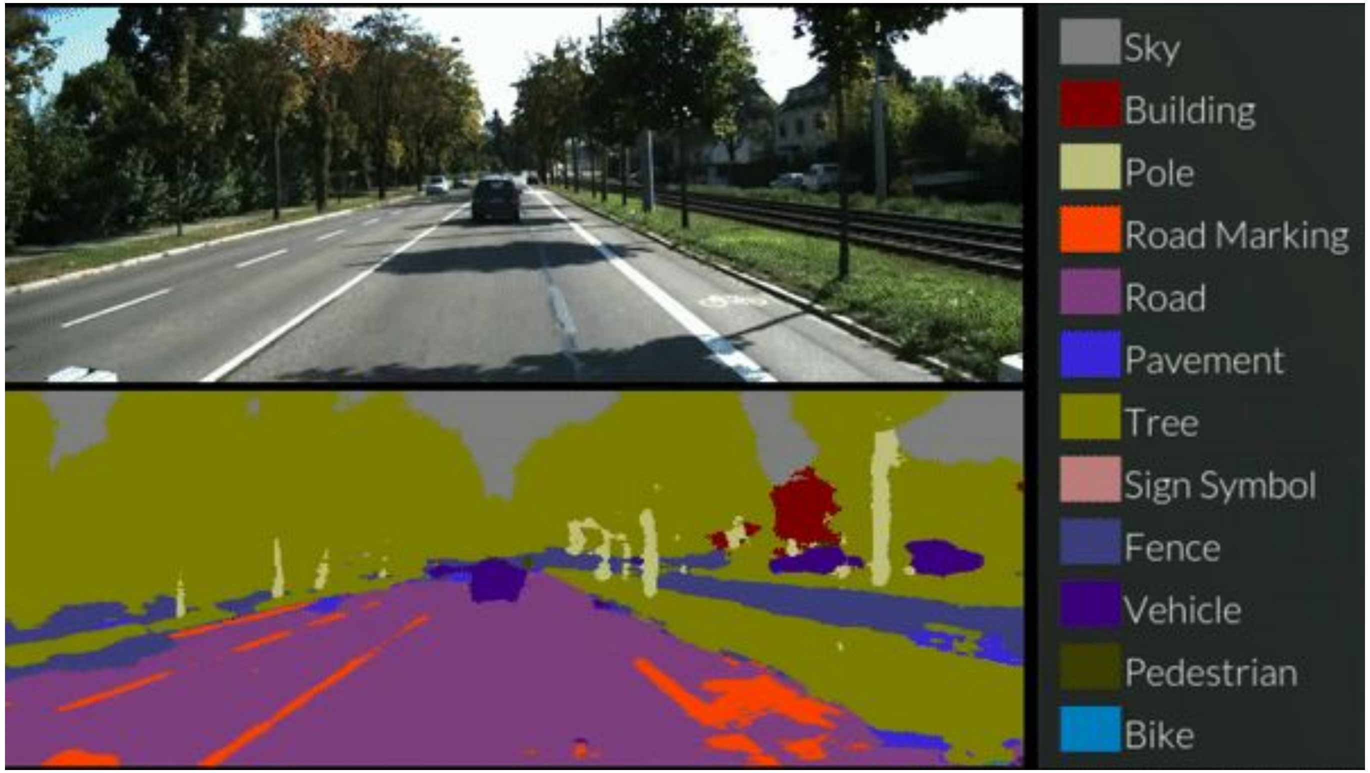


Object detection aims to locate and classify objects within an image.

- **Input:** Image (e.g., 640×640×3)
- **Output:** Bounding boxes, class labels, and confidence scores (e.g., [x1,y1,x2,y2], “car”, 0.92)

https://developers.arcgis.com/python-2-3/guide/yolov3-object-detector

2D Image - Semantic segmentation



Semantic segmentation aims to assign a class label to each pixel in an image.

- **Input:** Image (e.g., 512×512×3)
- **Output:** Pixel-wise segmentation mask (e.g., a 512×512 mask of class IDs)

https://medium.com/analytics-vidhya/introduction-to-semantic-image-segmentation-856cda5e5de8

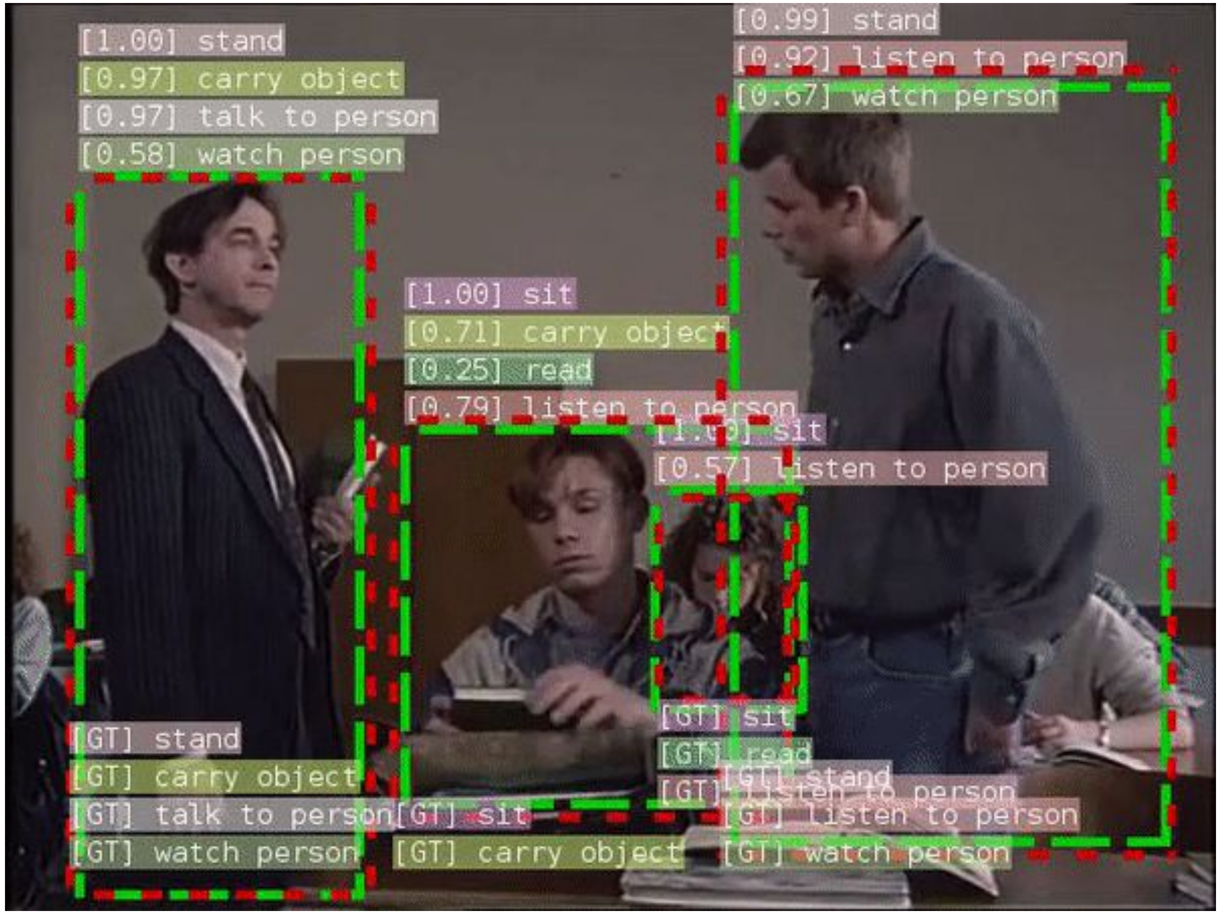
2D Image - Pose estimation

Pose estimation aims to localize keypoints of a person in an image.

- **Input:** Image (e.g., 256×256×3)
- **Output:** Keypoint coordinates with confidence scores (e.g., (x1,y1,c1),…)

https://github.com/ViTAE-Transformer/ViTPose

2D Video - Action recognition

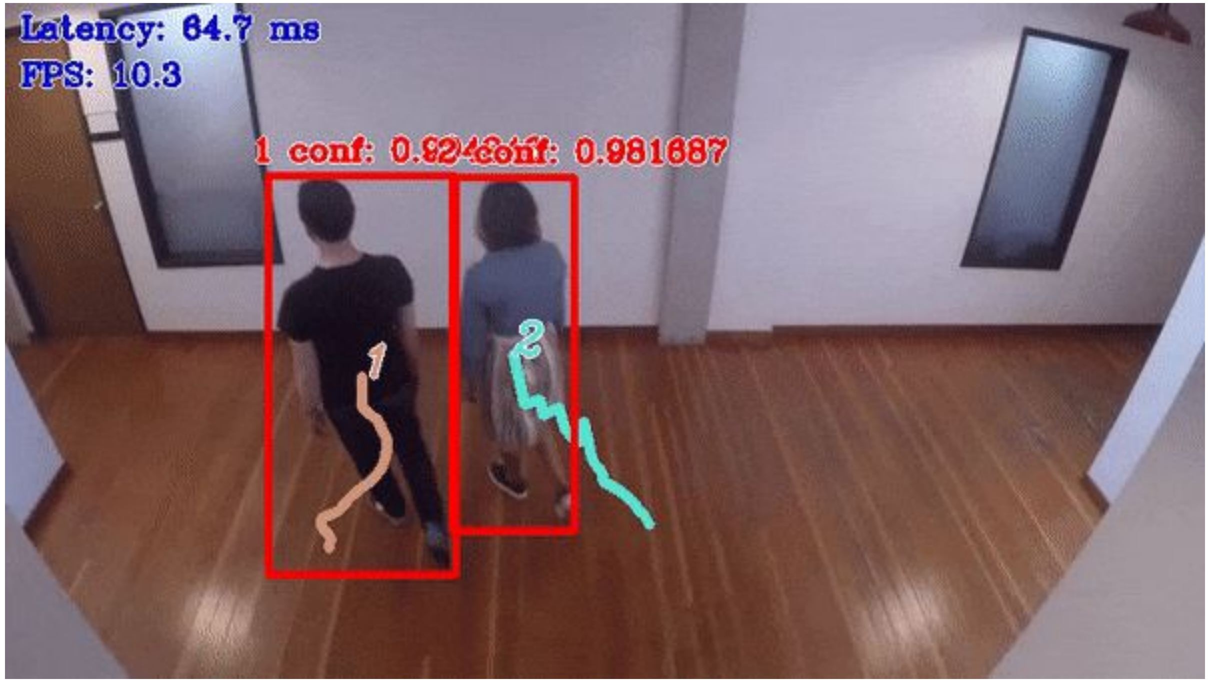


2D video action recognition aims to identify the action occurring in a video clip.

- **Input:** Video clip (e.g., T frames of size 224×224×3)
- **Output:** Probability distribution over action classes (e.g., [0.87, 0.10, 0.03])

https://github.com/facebookresearch/SlowFast

2D Video - Tracking



2D video tracking aims to follow objects across frames in a video.

- **Input:** Video clip (e.g., T frames of size 640×640×3)
- **Output:** Object trajectories over time (e.g., per-frame bounding boxes with IDs)

https://docs.openvino.ai/2023.3/omz_demos_pedestrian_tracker_demo_cpp.html

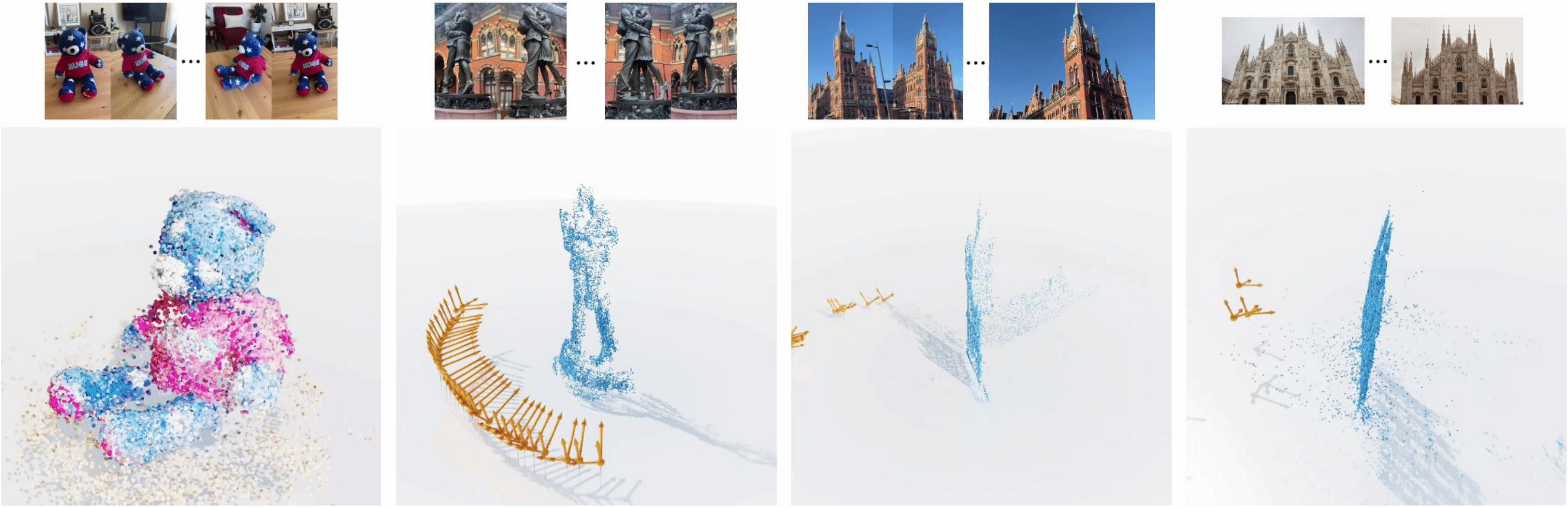
2D Video - Video segmentation

Video segmentation aims to assign a class label or object mask to each pixel across video frames.

- **Input:** Video clip (e.g., T frames of size 512×512×3)
- **Output:** Per-frame segmentation masks (e.g., T masks of size 512×512)

https://ai.meta.com/blog/segment-anything-model-3

3D Data - 3D reconstruction



3D reconstruction aims to recover a 3D shape or scene from input observations.

- **Input:** Multi-view images (e.g., T frames of size 512×512×3)
- **Output:** 3D geometry representation (e.g., point cloud, mesh, or voxel grid)

https://vggsfm.github.io

3D Data - Depth estimation

Depth estimation aims to predict the distance from the camera to each pixel in the scene.

- **Input:** Image (e.g., 512×512×3)
- **Output:** Depth map (e.g., a 512×512 matrix of depth values)

https://learnopencv.com/depth-anything

Multimodal Input - VQA

User Can you explain this meme?

GPT-4 This meme is a joke that combines two unrelated things: pictures of the earth from space and chicken nuggets.
The text of the meme suggests that the image below is a beautiful picture of the earth from space. However, the image is actually of chicken nuggets arranged to vaguely resemble a map of the world.
The humor in this meme comes from the unexpected juxtaposition of the text and the image. The text sets up an expectation of a majestic image of the earth, but the image is actually something mundane and silly.

Visual Question Answering (VQA) aims to answer a natural-language question based on an image.

- **Input:** Image + text question (e.g., 224×224×3 + "What is the boy holding?")
- **Output:** Answer in text form (e.g., "a ball")

Achiam, J., Adler, S., Agarwal, S., Ahmad, L., Akkaya, I., Aleman, F. L., ... & McGrew, B. (2023). Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*.

Multimodal Input - Text to image generation

a space elevator, cinematic scifi art | A cheeseburger with juicy beef patties and melted cheese sits on top of a toilet that looks like a throne and stands in the middle of the royal chamber. | a hole in the floor of my bathroom with small gremlins living in it | a small office made out of car parts | This dreamlike digital art captures a vibrant, kaleidoscopic bird in a lush rainforest. | human life depicted entirely out of fractals | an origami pig on fire in the middle of a dark room with a pentagram on the floor

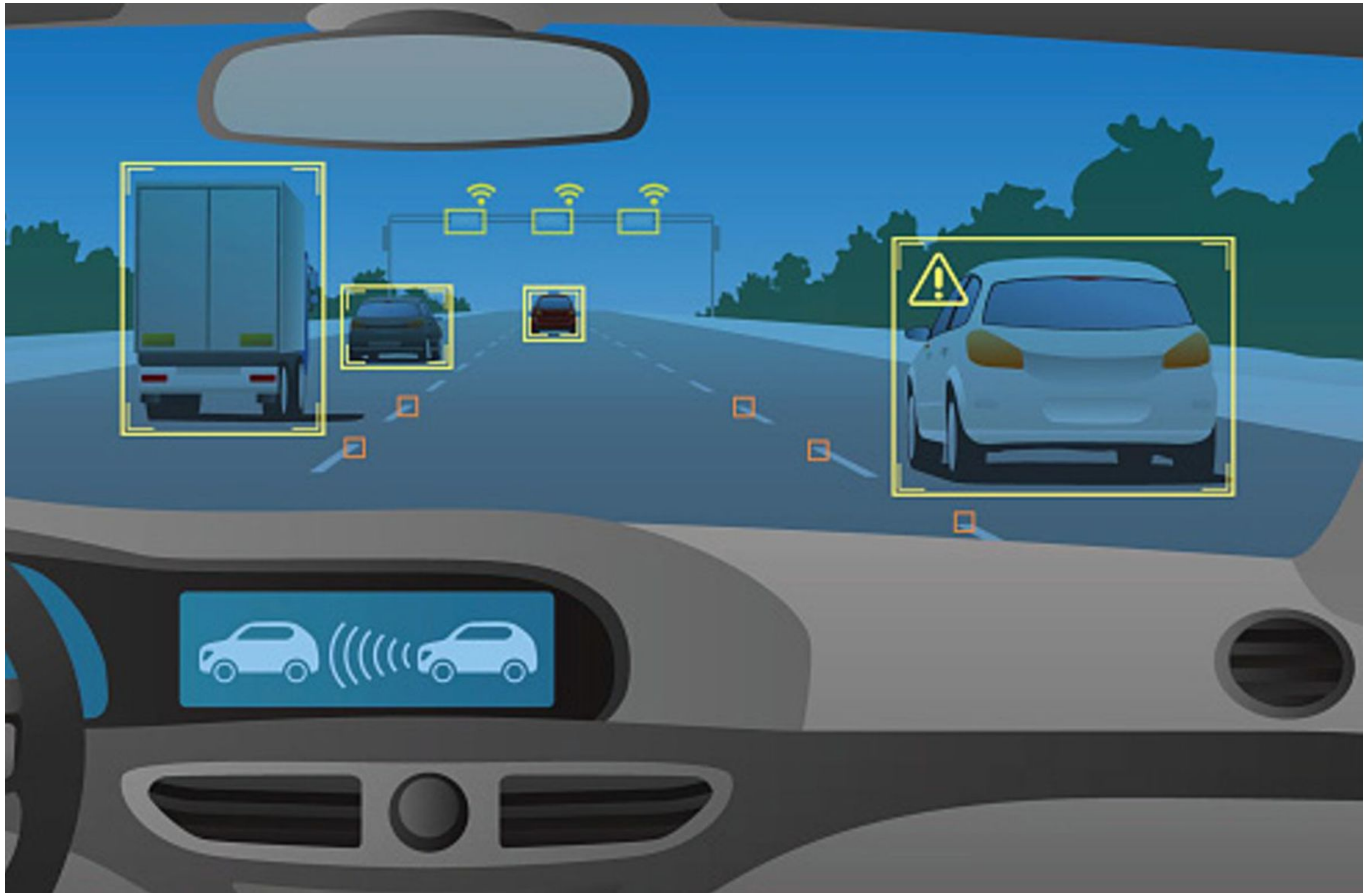
Text-to-image generation aims to create an image that matches a given text description.

- **Input:** Text prompt (e.g., “a cat sitting on a chair”)
- **Output:** Synthesized image (e.g., 512×512 RGB image)

bristol.ac.uk

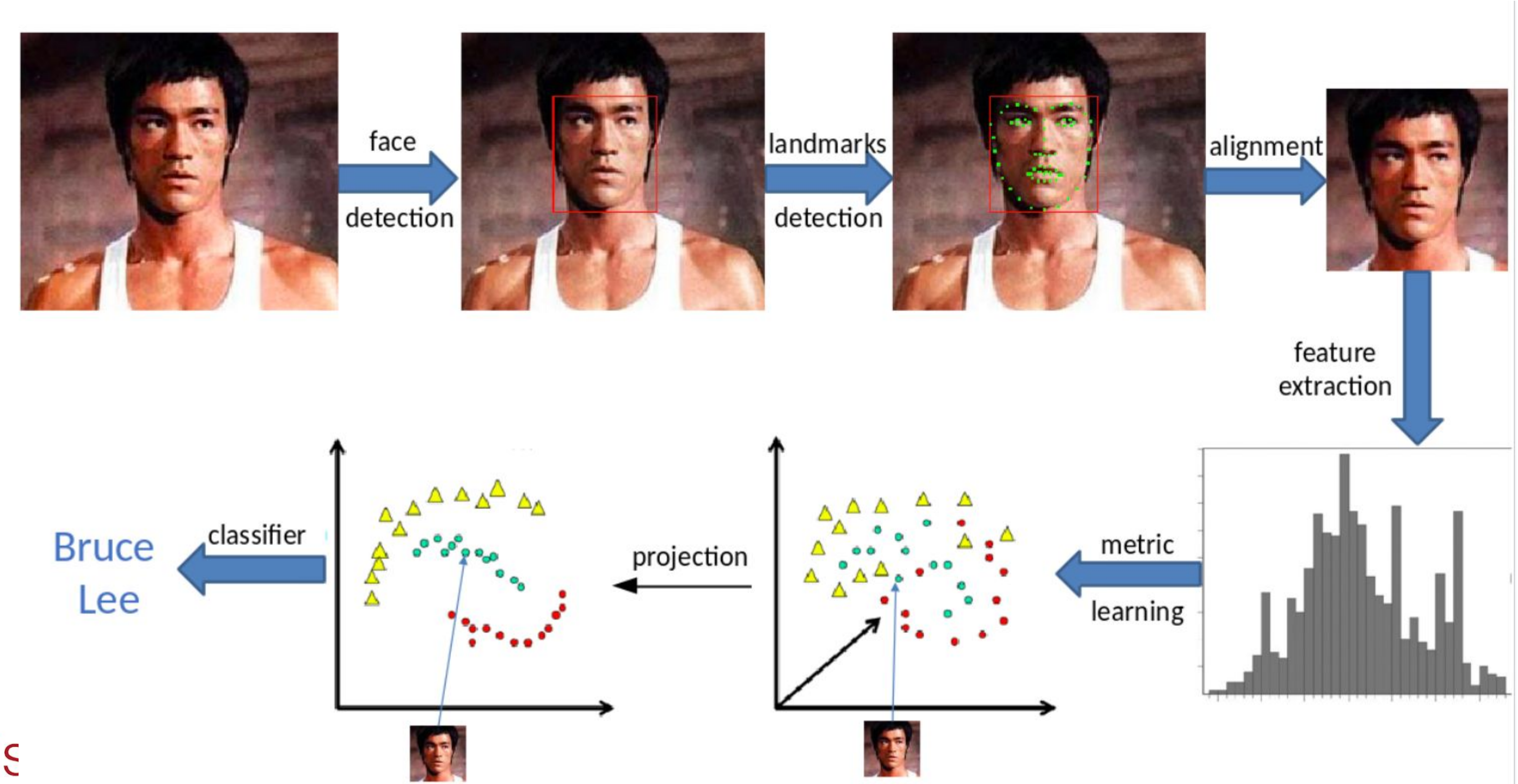
Esser, P., Kulal, S., Blattmann, A., Entezari, R., Müller, J., Saini, H., ... & Rombach, R. (2024, July). Scaling rectified flow transformers for high-resolution image synthesis. In *Forty-first international conference on machine learning*.

Application - Autonomous Driving



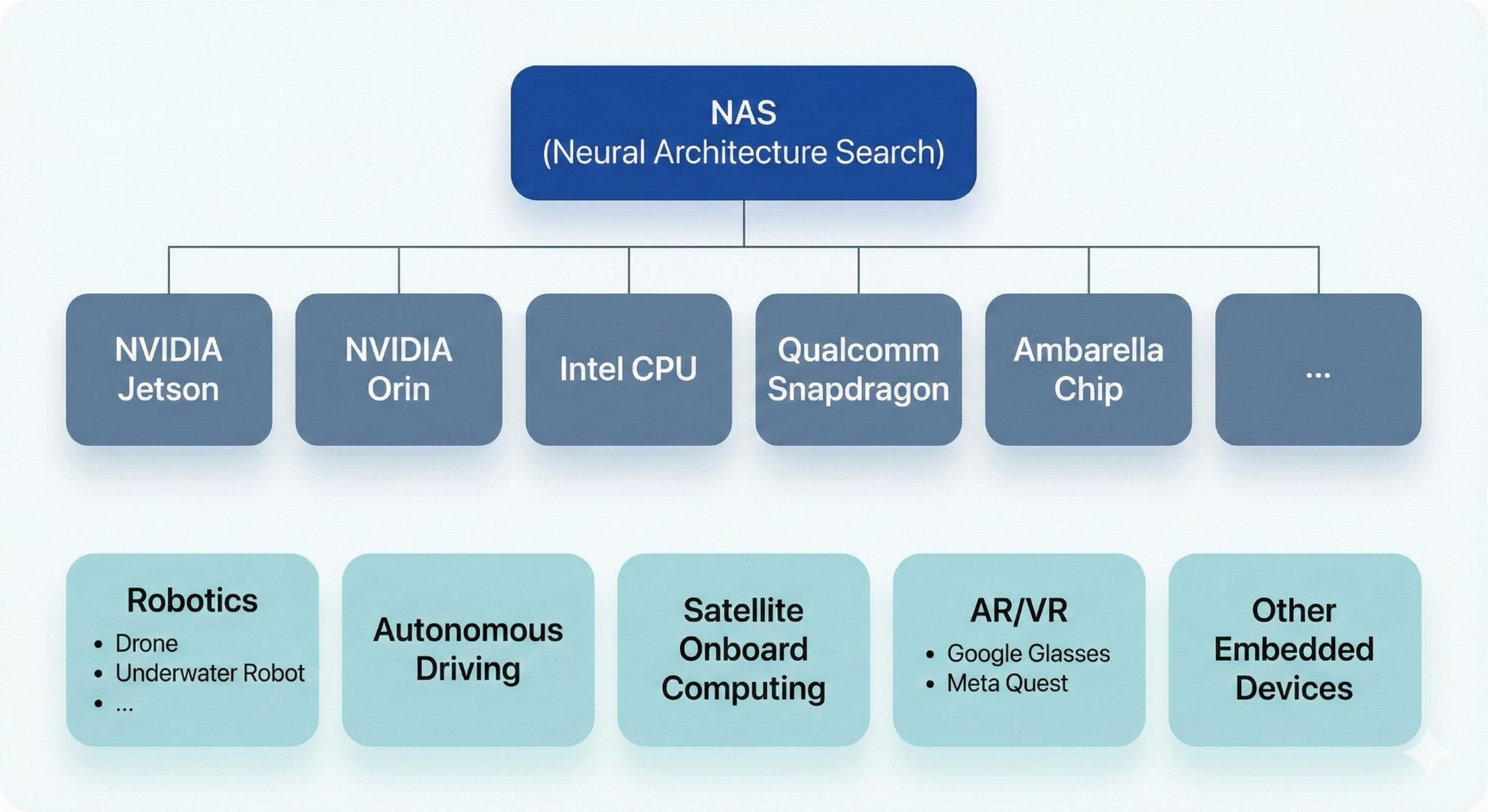
https://wvutoday.wvu.edu/stories/2017/03/06/wvu-researcher-to-study-computer-vision-related-image-recognition

Application - Face Recognition



bris

Figure Generation - Nano Banana in Gemini



AI Try On - Nano Banana in Gemini

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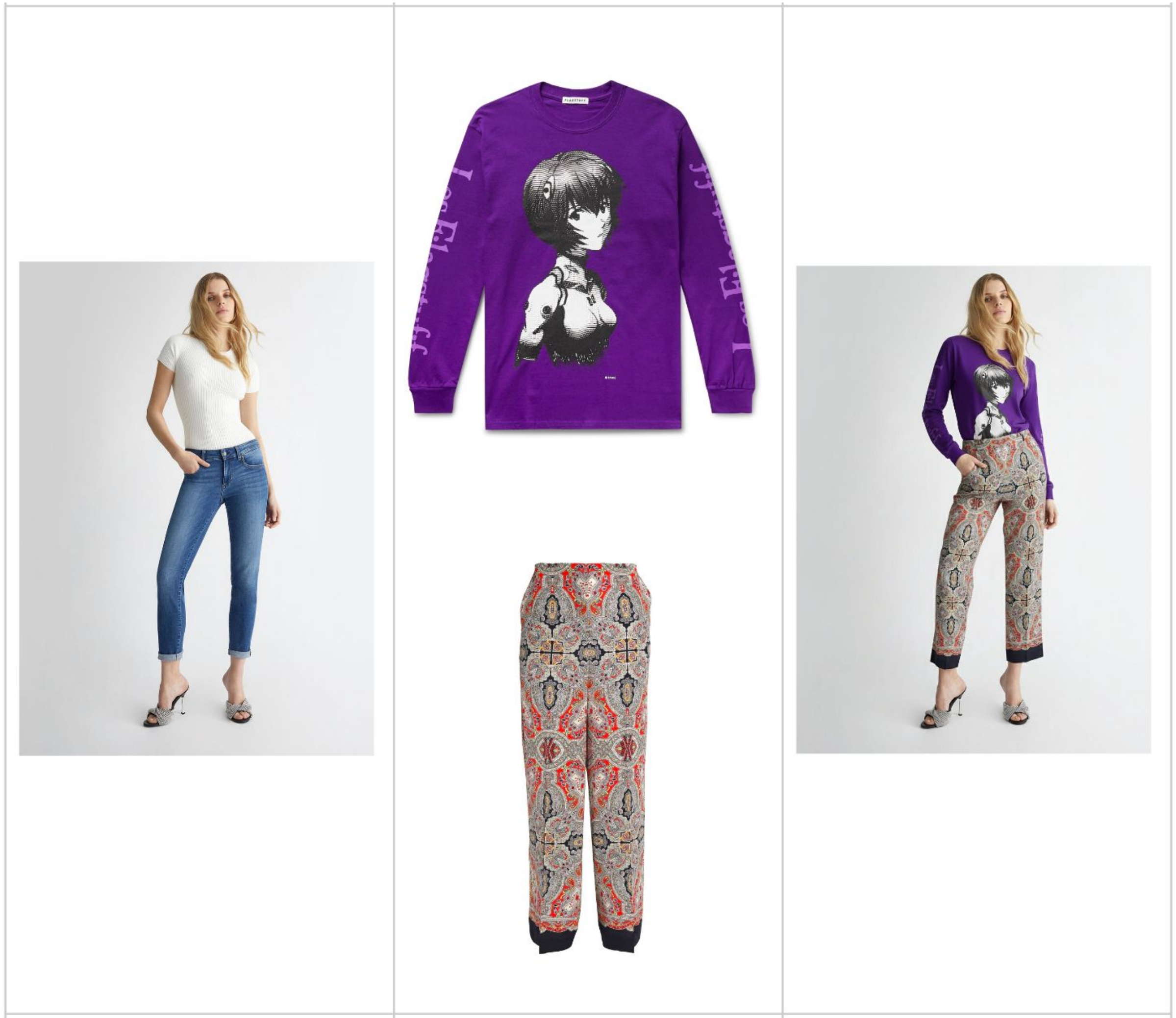

Multi-turn editing - Nano Banana in Gemini

https://blog.google/products/gemini/updated-image-editing-model/

Mix up designs - Nano Banana in Gemini